



CAMERA CULTURE GROUP
MIT MEDIA LAB

STARTUP GUIDE

SPAD Kit

MEMBERS WEEK · OCTOBER 2026

WELCOME



PARAGRAPH FROM RAMESH
TO BE INSERTED

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Documentation, code and datasets at

[Insert new site link]

Measuring the time light takes to travel

- 
- 2008 ● Camera Culture founded
MIT MEDIA LAB
 - 2009 ○ Looking around corners
ICCV · MARR PRIZE HM
 - 2012 ○ Hidden 3D shape from a single wall
NATURE COMMUNICATIONS
 - 2013 ○ Femto-photography
SIGGRAPH · TEST OF TIME
 - 2015 ○ Tags located behind turbid layers
NATURE COMMUNICATIONS
 - 2016 ○ Reading a closed book
NATURE COMMUNICATIONS
 - 2018 ○ Photography through fog
ICCP
 - 2023 ○ Two-bounce transients
CVPR
 - 2024 ○ PlatoNeRF
CVPR
 - 2025 ○ Navigating by hidden objects
ICRA
 - 2026 ● Hidden objects with consumer LiDAR
NATURE
 - 2026 ● DENALI dataset
CVPR

What we use it for

01 Non-line-of-sight

Seeing objects hidden around corners from light that bounces off a wall.

AROUND CORNERS · NATURE 2026 · DENALI

02 3D vision

Recovering full geometry, including mirrors and glass, from low-cost depth sensors.

PLATONERF · BLURRED LIDAR

03 Through scattering

Imaging through fog, tissue and layered materials by sorting photons by arrival time.

FOG · CLOSED BOOKS · SNOWPACK

04 Low-light

Perception from individual photon counts when there is almost no light.

05 High-speed

Capturing motion faster than conventional cameras can follow.

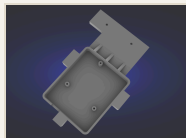
FEMTO-PHOTOGRAPHY

The kit



ST VL53L8CH

P-NUCLEO-53L8A1 expansion and development board



Printed mount

3D-printed housing for the ST sensor



USB cable

USB-A to Mini-B, connects the NUCLEO board to your laptop



Your computer

Mac, Windows and Linux are all supported

Put it together

1

STEP 1 · IMAGE + INSTRUCTIONS

2

STEP 2 · IMAGE + INSTRUCTIONS

3

STEP 3 · IMAGE + INSTRUCTIONS

4

STEP 4 · IMAGE + INSTRUCTIONS

5

STEP 5 · IMAGE + INSTRUCTIONS

6

STEP 6 · IMAGE + INSTRUCTIONS

Set up your laptop

Open the Terminal app and type each line, pressing Return after each one. You only do this once.

1 · INSTALL MINICONDA

Download and install it from

docs.conda.io/miniconda

2 · DOWNLOAD THE SOFTWARE

```
git clone https://github.com/NikhilBehari/spad_capture.git
cd spad_capture
```

3 · INSTALL IT

```
conda env create -f environment.yml
conda activate spad_capture
pip install -e .
```

4 · PLUG IN AND SET UP THE BOARD

Connect the sensor to your laptop with the USB cable, then:

```
spad st flash
```

See the photons

With the sensor plugged in, type:

```
spad st capture --viz
```

Then open this address in your web browser:

127.0.0.1:8888



DASHBOARD SCREENSHOT

Wave a hand in front of the sensor. The peak moves as the distance changes. Your capture is saved in the outputs folder.

What's next

Software and documentation

Every setting, zone selection and data format.

github.com/NikhilBehari/spad_capture

Datasets

DENALI and other captures from these sensors.

photons.media.mit.edu

CVPR 2027 challenge

Capture data with your kit and contribute to the benchmark.

photons.media.mit.edu



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